Introduction

The rapid evolution of the job market, particularly in the Information and Communication Technology (ICT) domain, has intensified the demand for effective interview preparation tools. Job seekers, especially in fields like artificial intelligence (AI), software development, and cloud computing, often face challenges in practicing realistic interviews due to limited resources and access to feedback. InterviewBot is a web-based platform designed to simulate virtual interviews using AI-driven question generation, text-to-speech capabilities, and performance analytics. This proposal defines the problem, outlines measurable objectives, establishes relevance to the ICT domain, and conducts a feasibility analysis. In Version 2 (V2) of the project, the system has been enhanced to include dynamic follow-up questions based on user responses, improving the realism and depth of the simulation compared to V1's static questioning approach. The project aims to bridge the gap between theoretical knowledge and practical interview skills, ultimately enhancing employability in the ICT sector.

Problem Statement

In the ICT domain, job candidates frequently struggle with interview preparation due to the lack of accessible, personalized simulation tools that mimic real-world technical and behavioral questioning. Recruiters report that up to 70% of candidates fail to demonstrate adequate communication and problem-solving skills during interviews, leading to prolonged hiring processes and mismatched placements. Existing platforms, such as traditional mock interview apps, often provide generic questions without tailoring to individual resumes or job descriptions, resulting in ineffective practice. This issue is exacerbated for entry-level ICT professionals, where technical interviews require domain-specific knowledge in areas like AI algorithms or cloud architectures. The problem is a need for an affordable, AI-powered tool that delivers realistic, adaptive interview experiences with immediate feedback to better prepare candidates and streamline recruitment.

Objectives

The project objectives are defined using the SMART framework to ensure clarity and achievability:

1. **Specific**: Develop a web-based platform that generates tailored interview questions based on user resumes and job descriptions, incorporating AI for dynamic follow-ups (enhanced in V2 to probe deeper based on previous answers).
   - **Measurable**: Achieve at least 85% relevance in generated questions, validated through user testing with 20 participants.
   - **Achievable**: Leverage open-source AI APIs like Google Gemini and existing web frameworks.
   - **Relevant**: Addresses ICT interview challenges, such as technical depth in AI/ML roles.
   - **Time-bound**: Complete prototype by December 2025.
2. **Specific**: Integrate text-to-speech and speech recognition for interactive simulations, with real-time feedback on user responses.
   - **Measurable**: Ensure <3-second response time for TTS and 90% accuracy in speech recognition.
   - **Achievable**: Use ElevenLabs API with gTTS fallback.
   - **Relevant**: Enhances user engagement in virtual ICT environments.
   - **Time-bound**: Implement by January 2026.
3. **Specific**: Generate performance summaries with scores on communication, confidence, and domain knowledge, exportable as PDFs.
   - **Measurable**: Validate scores against manual reviews, targeting 80% alignment.
   - **Achievable**: Use pdflatex for PDF generation.
   - **Relevant**: Provides actionable insights for ICT career development.
   - **Time-bound**: Finalize by February 2026.
4. **Specific**: Incorporate session recording and ethical data handling features.
   - **Measurable**: Achieve 100% user consent compliance in tests.
   - **Achievable**: Via MediaRecorder API.
   - **Relevant**: Aligns with ICT privacy standards.
   - **Time-bound**: Integrate by November 2025.

Relevance to ICT Domain

InterviewBot is deeply rooted in the ICT domain, specifically AI and web development. It leverages AI for natural language processing (NLP) to generate context-aware questions, reflecting trends in machine learning applications for human-resource tech. According to an IEEE report, AI-driven tools are transforming recruitment by automating 60% of initial screening processes (IEEE, 2024). The platform's use of web technologies (Flask, Three.js) aligns with cloud-based ICT solutions, enabling scalable virtual simulations similar to those in remote collaboration tools.

In V2, the addition of adaptive questioning mirrors advanced AI systems like conversational agents in chatbots, addressing ICT challenges in personalized user experiences. The project contributes to ICT by promoting AI ethics (e.g., data privacy) and accessibility in education, as highlighted in ACM studies on AI in career development (ACM, 2025). Overall, InterviewBot advances ICT by combining AI, multimedia processing, and web deployment to solve real-world problems in talent acquisition.

Feasibility Analysis

Technical Feasibility

The project requires tools like Python (Flask) for backend, JavaScript (Three.js) for frontend, Tesseract for OCR, Google Gemini for AI, and ElevenLabs for TTS. These are suitable due to their maturity and compatibility—Flask enables rapid API development, while Gemini provides robust NLP without custom training. V2's follow-up logic was implemented by extending the Gemini prompt, proving feasible with minimal code changes. Resources include a local development machine and free tiers of APIs. Justification: These technologies are open-source or low-cost, with extensive documentation (Flask Docs, 2023).

Economic Feasibility

Estimated costs are low: $0 for open-source tools (Flask, Tesseract); $10/month for Gemini/ElevenLabs APIs during development (free tiers suffice for testing); $5/month for Render hosting. Total budget: <$50. Affordable within student constraints, with no hardware purchases needed.

Ethical Considerations

Potential issues include data privacy (user resumes, recordings) and AI bias in question generation. Mitigation: Obtain explicit consent in rules.html, anonymize data, and use diverse prompts to reduce bias. Complies with GDPR principles for user consent and data minimization.

Market/User Needs Analysis

The target market includes job candidates (primary users), recruiters, educational institutions, and career coaches in the ICT sector. Market analysis reveals high demand: A Gartner report (2024) indicates 75% of ICT job seekers desire AI-based preparation tools, while a McKinsey study (2024) notes recruiters spend 23% more time on interviews due to skill mismatches. User needs include realistic practice (LinkedIn, 2023: 68% feel underprepared) and feedback (IEEE, 2024: 70% want personalized insights). Educational users seek scalable tools (ACM, 2025: Bootcamps report 90% demand for simulation platforms). Sources confirm the gap: Glassdoor (2023) highlights costly alternatives like Interviewing.io, underscoring the need for affordable solutions like InterviewBot.

Existing solutions like Pramp offer peer-based mocks but lack AI tailoring (Pramp Case Study, 2023). Interviewing.io provides expert feedback but is premium-priced (IEEE, 2024). InterviewBot's novelty lies in its free, AI-driven adaptation (V2 follow-ups) and 3D immersion, improving over static tools. A 2025 ACM review of AI in HR notes that adaptive systems increase candidate success by 40%, supporting InterviewBot's approach as an improvement in accessibility and depth.

Conclusion

InterviewBot addresses a critical ICT problem by providing an AI-powered interview simulator. With SMART objectives, strong ICT relevance, and feasible implementation, the project promises impactful contributions. V2 enhancements demonstrate iterative progress, ensuring a robust solution for stakeholders.

(References:

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